

Curiosity-Inspired Learning: Insitu versus Post-Event Approaches to Recall and Reflection

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ABSTRACT

We often get questions about the processes and things that we observe in our surroundings, but there exists no practical support for exploring these questions. Exploring curiosity can lead to learning new science concepts. We propose using a post-event recall and reflection approach to support curiosity-inspired learning in everyday life. This approach involves capturing contextual cues during the curiosity moment with wearables that can capture these contextual cues in daily life, and later using them for recall and focused reflection. Firstly, we conducted a preliminary study to explore different cues and their effectiveness in recalling these curiosity moments. Further, we conducted a virtual study to evaluate the amount of exploration through post-event recall and reflection and compared it with insitu recall and reflection. Results show a significant increase in questions and reflections made with the post-event recall and reflection approach, providing evidence for better learning outcomes from everyday curiosity.

CCS CONCEPTS

• **Curiosity** → **Learning**; • **Wearables** → *Capture context*; • **Recall and Reflection**;

KEYWORDS

Cued Recall; Reflection; Curiosity; Moment Capture; Contextual Cues; Wearables

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1 INTRODUCTION

Curiosity is the innate drive and intrinsic motivation for learning and acquiring new information [3]. Reflecting on moments in which curiosity strikes and exploring them can be a pragmatic medium to learn things and to find even real-life applications for previously acquired knowledge. Additionally, curiosity exploration can lead to learning new science concepts [21]. Such a learning experience would be more enjoyable and self-motivated compared to traditional formal learning. In our daily life, we often have curiosity questions, but, most of the time, we do not explore them due to time constraints, the task at hand, or lack of technological resources at hand like the internet, a computer, etc. Even when people have the resources available, exploring the curiosity comes at the cost of the lack of presence in the task which the person was involved in the first place. While curiosity can be the most natural and practical motivation for scientific learning, clearly there is a lack of technological support to overcome the constraints posed [1]. These issues can be worked by availing time and resources for curiosity exploration. One possible way would be simply availing the time and resources needed when having the curiosity moment (insitu). Another way would be having curiosity exploration when one has the time and resources (post-event). Curiosity in daily life occurs randomly, while ensuring having time and resources all in place to explore is more of a scheduled event. In order to fill this gap, a random curiosity event has to be linked with having the time to explore it. Recalling and recreating the moment of curiosity at a scheduled time (when one is available) of exploration can help bridge the gap. Contextual cues can help in memory retrieval and in effectively recalling such moments [12]. Cues are moment associated information that assists in the retrieval of memories.

There are numerous lifelogging applications and ubiquitous systems which also capture environmental cues to recall the moment later, but these systems are merely aimed at logging life events and capturing memories to allow people to revisit them later. These applications fail to explore practical purposes [18] such as the learning embedded in these everyday life moments, the opportunity to revisit logged moments for the purpose of reflection, etc. In this paper, we present and evaluate an approach to supporting curiosity-based learning through the capture of contextual cues and the use of these cues for post-event recall and reflection.

Smart wearables, ubiquitous technology, have the potential to support curiosity-based learning in everyday life. Wearables, with

their advanced sensors, can support the capturing of contextual cues such as the image of the environment, the time and location of the event, etc., and can be carried to all of the environments that the user visits. The post-event recall and reflection approach involves using a system that allows the capturing of contextual cues to be used later (post-event) by the user to revisit curiosity moments for recalling the question and then reflecting. We evaluate this approach against the insitu (in the moment) recall and reflection approach. Comparing it to the insitu approach allows us to evaluate if a similar level of exploration can be achieved by separating the curiosity moment and the recall and reflection.

Firstly, we conducted a preliminary study to explore what cues are important to people for recalling their moments of curiosity. We evaluated eight different contextual cues (action, emotion, image, location, object, people, physiological, and temporal) for their helpfulness in recalling. Then, we conducted an online study to evaluate the approach of moment capture and post-event recall and reflection in terms of the quantity of questions asked and the quality of reflection made, with a baseline approach of insitu recall and reflection. This study was conducted in a virtual setting using a teleconferencing software, and 360-degree images were used as the virtual environments to explore. We analyzed the data collected from 30 participants and present the results. Finally, we discuss the implication of the results and future work.

2 BACKGROUND AND RELATED WORK

Curiosity can be broadly defined as the exploratory behaviour which brings the desire to acquire new knowledge [11]. There are two major types of curiosity: perceptual curiosity (related to sensory stimuli) and epistemic curiosity (drive to fill information gap). *Epistemic curiosity* relates to the drive to know more and fill in the information gap [2], and is therefore interesting to explore for learning outcomes.

There are a few papers in the past that have proposed systems to support the capturing of moments and revisiting them, but they do not support curiosity based learning directly. For example, Li et al. [10] proposed a learning log system called *Scroll* which supports the capture of a user's learning logs and allows for the reviewing and reflecting of their old related logs. In another paper by Hinckley et al. [9], they proposed a system to support note-taking for important ideas. Technologies as an aid for memory has a long history, and lifelogging is one of its outcomes. Multiple lifelogging support technologies are available for capturing moments [15, 19]. While some systems support lifelogging through wearables, there are many mobile phone based applications as well [5, 6]. *Journaly* is an example of a moment capturing interface that allows the user to capture moments of interest by gathering information related to the moments like notes, pictures, location, and other data together [6]. While there are numerous such systems to support journaling in capturing the moments of interest, there are rarely any systems that exploit this capability for the purpose of learning from everyday curiosity. These applications lack associating the capture of moments to learning or curiosity, and do not have the support for post-event exploration. Instead, they aim at creating memories, just like having a photo album with contextual texts.

While curiosity in everyday life has the potential for learning, curiosity itself is not enough to build learning outcomes [1], due to constraints like time, resources, the task at hand, etc. Thus, there is a need for technology to support and channel curiosity to the science learning of interest and relevance. Arnone et al. [1] proposed a model of curiosity, interest, and engagement for new media technology-pervasive learning environments.

We root our approach in the conclusion from Ferreira [8], where two ways of reflecting were acknowledged: reflect-in-action and reflect-on-action. *Reflect-in-action* implied reflecting on our thoughts at the same time when performing an action, whereas *reflect-on-action* implies the reconstruction of the action, once it has concluded, then reflecting on our thoughts and learning from the experience.

3 RESEARCH QUESTION

The research question we want to explore is: Does *curiosity exploration* vary when people explore during the curiosity event vs after the curiosity event?

We measure curiosity exploration through the number of curiosity-inspired questions asked and the number of times participants reflected on these curiosity moments.

4 PRELIMINARY STUDY

A preliminary study was conducted to evaluate usefulness of cues in recalling. Investigating the helpfulness of cues in recalling is important for understanding what set of information would help recalling the moment and hence would affect the type of technology device involved to capture those contextual cues. Also, this preliminary study will provide an initial result on how well people can recall, therefore setting the ground for using the recall and reflection approach.

This study was approved by the university ethics board and had a total of 11 participants, consisting of 6 female and 5 male. The participants were also university students. We investigated cued recall, i.e. memory retrieval, with the aid of cues. Eight different cues were evaluated: action, emotion, image, location, object, people, physiological (heart rate), and temporal cues.

The study was divided into two sessions, and the participants' consents were obtained prior to the study. In the first session participants were assigned a few walk-able locations to explore and were told to ask five questions that piqued their curiosity. The participant was shadowed by a researcher who recorded the entire session. The participants were wearing an ion snap cam [20] (to capture the first-person view of the moment of curiosity) as a wearable smart clip and an Empatica E4 wristband [7] (to capture physiological data). Then, the second session was conducted after an approximate interval of a day. In this second session, participants were asked to use the recall interface (Figure 1), i.e. to use the cues from the interface to recall the moment and note their question in the interface. The recall interface consisted of the eight cues in the form of tiles. Users could reveal each cue by clicking on corresponding cue's tile.

The image, object, location, temporal, and people cues were obtained from the smart clip. The action and emotion cues were obtained from the video recordings during the shadowing. The final emotion cue was obtained by processing the participant's face

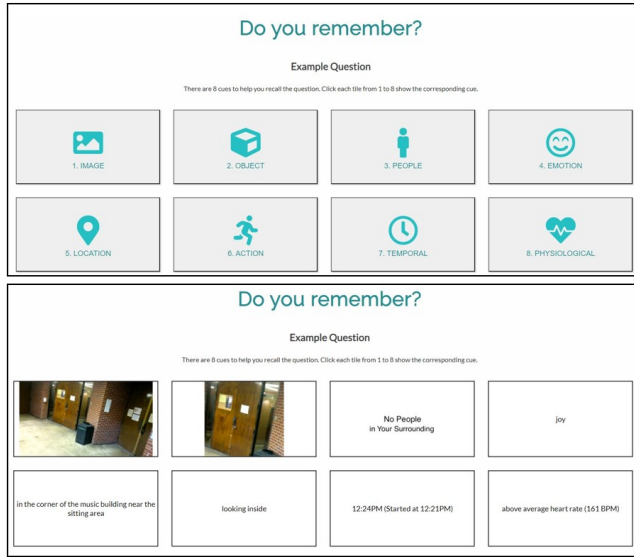


Figure 1: Recall interface: the interface used in the preliminary study to evaluate cues helpfulness in recalling. The top part shows the page with eight cues listed in the form of tiles, and the bottom part shows the interface when all the tiles have been turned.

(during asking question) through Affectiva, an emotion measurement technology, which uses human facial data to output emotion. Empatica E4 captured the physiological cues.

Analyzing the response to "At which cue were you the most confident?", we found that image (62% of the times), object (24% of the times), and location (11% of the times) were rated as the most important cues in that order. Another measure used was self-reported ratings of the cues' helpfulness in recalling the question. A seven-point scale was used, with 1 as 'extremely unhelpful' and 7 as 'extremely helpful'. Average rating for each of the cues respectively are Image - 6.2, Object - 5.6, Location - 4.9, People - 2.6, Action - 2.4, Temporal - 2.2, and Emotion - 1.7. Similar observations have been made in prior literature also [10]. For the accuracy of recall, the recalled questions were compared to original questions. Questions that matched exactly or matched in overall meaning, with some changes in detail or structure of sentence, were marked as correct. Participants were able to recall their questions 78% of the time.

5 ONLINE STUDY

We designed the final study to compare post-event recall and reflection with insitu recall and reflection for learning outcomes. Using the results from the preliminary study, we designed an interface with cues to help in recall. This study was approved by our university ethics board and was conducted through a teleconferencing software. A total of 30 participants were recruited through the university recruitment pool and through advertisement in the department. The participants were compensated with course credit for their time and effort. Signed consent forms were obtained prior to starting the study.

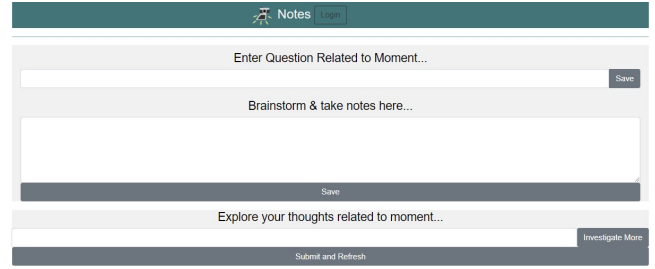


Figure 2: Notes interface: the interface used in the insitu condition for noting down questions, reflections and exploring online

5.0.1 Protocol. The entire study session was recorded. The study protocol required participants to explore two virtual environments in each condition. Virtual environments of similar qualities were chosen - one was a street view and the other a theme park view. These two types were chosen to have variation in environment type. The participants' task was to count the number of garbage bins in each environment. At the same time, they were asked to explore the environment and ask scientific questions that piqued their curiosity. The term *scientific question* was used to refrain people from asking simple what, who, and where questions. In the context of our study, a *scientific question* is defined as a question that related things and processes observed in the environment to science. The task of counting garbage bins was chosen as a distraction, to represent the idea of being involved in a task in daily life while observing the environment. Further, participants were asked to reflect upon their questions and explore them. *Reflection* was explained as thoughts that come to the mind concerning the curiosity moment, the question or the solution to the question, etc. Pre-downloaded 360-degree images were used as the virtual environment for close to real-life environment representation. In both conditions, the task of exploring the virtual environment, reflecting on the curiosity moment, and investigating were not time-constrained. Participants were asked to complete the task and explore as much as they wanted. Finally, after completing both conditions, participants were interviewed about their experience with the two approaches.

5.0.2 Study Design. A within-subjects study design was used, with the immediacy of reflection as the independent variable. The conditions were counterbalanced. The study was comprised of two conditions - *insitu recall and reflection* and *post-event recall and reflection*.

In the *insitu recall and reflection condition*, whenever participants had a question while exploring the virtual environment, they went to the *Notes* interface (figure 2) in the moment of curiosity, and took note of their question and reflection. The *Notes* interface allowed them to note and save their question, reflection, and to explore their questions in Google using the 'investigate more' features available in the interface. The 'investigate more' button navigated participants to Google with the question entered as the search term. In this condition, participants explored and reflected at the moment of curiosity.

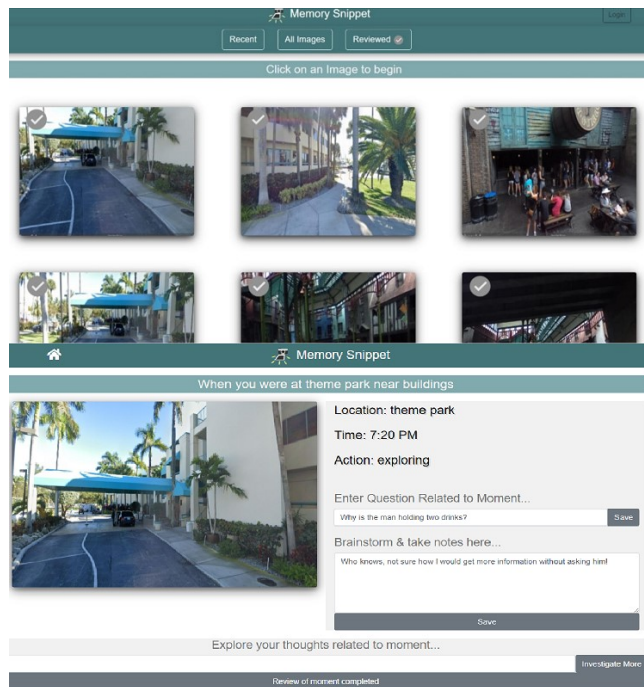


Figure 3: Memory Snippet: the interface used in the post-event recall and reflection condition for recalling, noting down questions and reflections, and exploring online. Top part of the image shows the home page, bottom page appears on clicking any image in the home page

For the *post-event recall and reflection condition*, whenever participants had a question, they clicked a *capture button*. This was a small button, present on the top left corner of the screen and placed on top of the virtual environment. When clicked, this button automatically captured environmental cues (a screenshot of the environment and a timestamp) to be used later for recall and reflection. After this, they continued the task, exploring the environment and asking questions, until they were done exploring. After completing the task of exploring and asking questions, an interval of 10 minutes or more (a distraction of filling a survey), was introduced.

Then, the participants were asked to use the *Memory Snippet* interface (figure 3) to recall their questions using the cues presented (captured through capture button click) and take notes of their question and reflection after recalling. This interface was designed using the results obtained from the preliminary study, i.e environmental cues like image (which also included object), location, and time were all used. The home page of the interface looked like a gallery of moments represented through images. The user could click on any image to get all the cues, enter questions and reflections, and explore. This interface was designed with image as the primary cue. Just like the notes interface, this interface also had the 'investigate more' feature. All the cues were manually entered by the researcher conducting the study. These cues were entered through Firebase into the interface. Also, all the information logged by the participants into the interface was saved in the Firebase database. In this condition, participants first completed the exploration and the task

at hand, and later recalled the moment and the question, followed by noting the question and the reflection.

5.0.3 Measures. Presence in the virtual environment was measured using the Igroup Presence Questionnaire (IPQ) with a 7-point likert scale [17]. Curiosity based exploration was measured through the number of questions asked and the quantity of reflection made. Quantity of reflection was measured by the notes taken down by participants through the *Notes* and *Memory Snippet* interface.

6 DATA ANALYSIS

A total of 30 participants took part in the study. The average age of the participants was 20.8 years. All of them were university students. There were 25 male and 5 female participants. For the IPQ scale, the overall average score was calculated for both of the conditions. Questions asked and the reflection made was obtained by data filled out and saved by the participants, through the interface for reflection (*Notes* interface in case of insitu condition and *Memory Snippet* interface in case of post event).

A dependent T-test was run to find any significant differences in terms of presence in the environment between the two conditions. The following data was collected by the *Memory Snippet* interface and *Notes* interface: the count of questions asked, the count of the number of times notes were taken, by each participant for each condition for reflecting upon questions. A dependent t-test, or a non-parametric alternative Wilcoxon signed-rank, was run for all the measures, depending on the normality of data. These tests were run on all of the data to find any significant difference between the insitu recall and reflection and the post-event recall and reflection condition. Tests were run with the immediacy of the recall and reflection as an independent variable.

7 RESULTS

As a manipulation check, we explored the difference in the presence of participants in both of the conditions. Results of paired sample t-test did not show any significant differences in terms of presence in the environment between both of the conditions. This implies that there is no evidence for differences in the presence of the virtual environment. The Wilcoxon Signed-Ranks Test indicated that the number of curiosity-inspired questions asked in the post-event recall and reflection condition (mean rank = 14.47) was statistically significantly higher than in the insitu recall and reflection condition (mean rank = 11.31); $Z = -2.17$, $p = .03$. To give an idea of the type of questions asked, we quote two asked questions; "Is construction roofing for sidewalks in cities effective", "What is butter beer made out of?" The results of the Wilcoxon Signed-Ranks Test indicated that the number of times people reflected on curiosity moments and their questions in the post-event recall and reflection condition (mean rank = 14.76) was statistically significantly higher than in the insitu recall and reflection condition (mean rank = 7.42); $Z = -3.19$, $p = .001$. Below is an example of a question and its related reflection note; Question: "Why did the sidewalk have that green stain?", Reflection notes: "It must have had something to do with the bushes next to it, but how that color was placed on the sidewalk I am not sure about. It must have been some sort of green fungus according to google."

8 DISCUSSION AND FUTURE WORK

Our results clearly indicate that the post-event recall and reflection approach, i.e. capturing contextual cues in the moment of curiosity, and later using those cues to revisit the curiosity moment, allows for more exploration. At the same time, it also encourages people to ask more questions. This approach has the potential to result in better exploration of one's curiosity, potentially due to the availability of unconstrained time when revisiting memories later, with the intention of reflection. Revisiting moments through captured cues can also increase one's focus on reflection as it becomes the main task, resulting in a deeper exploration and more meaningful process of learning. User initiated capture helped in efficiency of recall, this was observed by Niforatos et al. [14]. The image was found to be the most useful cue for recalling (in preliminary study) also, a few participants in the online study mentioned that they did not observe other cues. Also, associating each curiosity moment with contextual cues added value for future visits. The insitu approach by allowing one to think during an event actually distracts one from the original event [4], thus might not be appropriate.

While the majority of participants asked more questions in the post-event recall and reflection condition, some participants did ask more in the insitu conditions as well. The insitu condition had its own benefits where being present in the environment itself allowed participants to revisit the environment as they are noting their question to reaffirm their thoughts [16] as cited in [13]. Also, being in the original moment of curiosity allows for direct thinking.

In the post-event recall and reflection approach, participants spent more time reflecting on their thoughts and explored their questions by using the 'investigate more' button and navigating to Google for more information, while 'investigating more' in insitu condition led to a distraction from the original task and break of flow. Overall, having a system that supports curiosity-inspired learning using the recall and reflection approach can increase learning outcomes.

Our results show that providing support for curiosity can motivate these exploratory behaviours. While the final study was conducted in a virtual environment, our preliminary study shows that these contextual cues can be easily captured by smart wearable devices in a real-life environment. A limitation that might appear in real life, is the accuracy of the cues captured through the wearable. This limitation can be overcome by using an action camera and wearable device, capturing first-person images, and having user initiated cue capture [14].

For future work, there could be an exploration on whether such a support has the potential to directly contribute towards one's curiosity or not. Further investigating the quality of reflection and the type of questions asked will allow an understanding on other outcomes using this approach. In our first study, we used a gap of around a day and a maximum of two days between the curiosity event and the recall, while in the online study we used a gap of 10 minutes. In the future, we might also need to explore, after what time interval, between the curiosity event and the recall, does the recall rate fall to a point where the benefit of this approach diminishes. While our current work focuses on everyday learning that can be used by all, in the future we would like to explore the use of such support by focusing on a specific educational space like STEM

education. Such an approach can be employed in school learning as well as in lab sessions by allowing students to capture the moment of curiosity in class and lab, and later revisit and explore it.

9 CONCLUSION

We performed initial investigation to compare insitu and post-event recall and reflection approach for curiosity exploration. Through the virtual prototype, we observed that this approach not only allows for a better reflection but also enables people to ask more questions than they would in the traditional in the moment note-taking approach. Also, people tend to explore their questions more when they use a system to support post-event recall and reflection, possibly due to enabling reflection to become the focus and availability of time without the disruption from other tasks. These initial results provide evidence for further exploration on the post-event recall and reflection approach to channel curiosity towards learning. Such a support system can be potentially used for formal as well as informal learning.

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